

Теория вероятностей
Индивидуальное домашнее задание
Вариант 9

1	2	3	4
+	+	+	+

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Проверил преподаватель

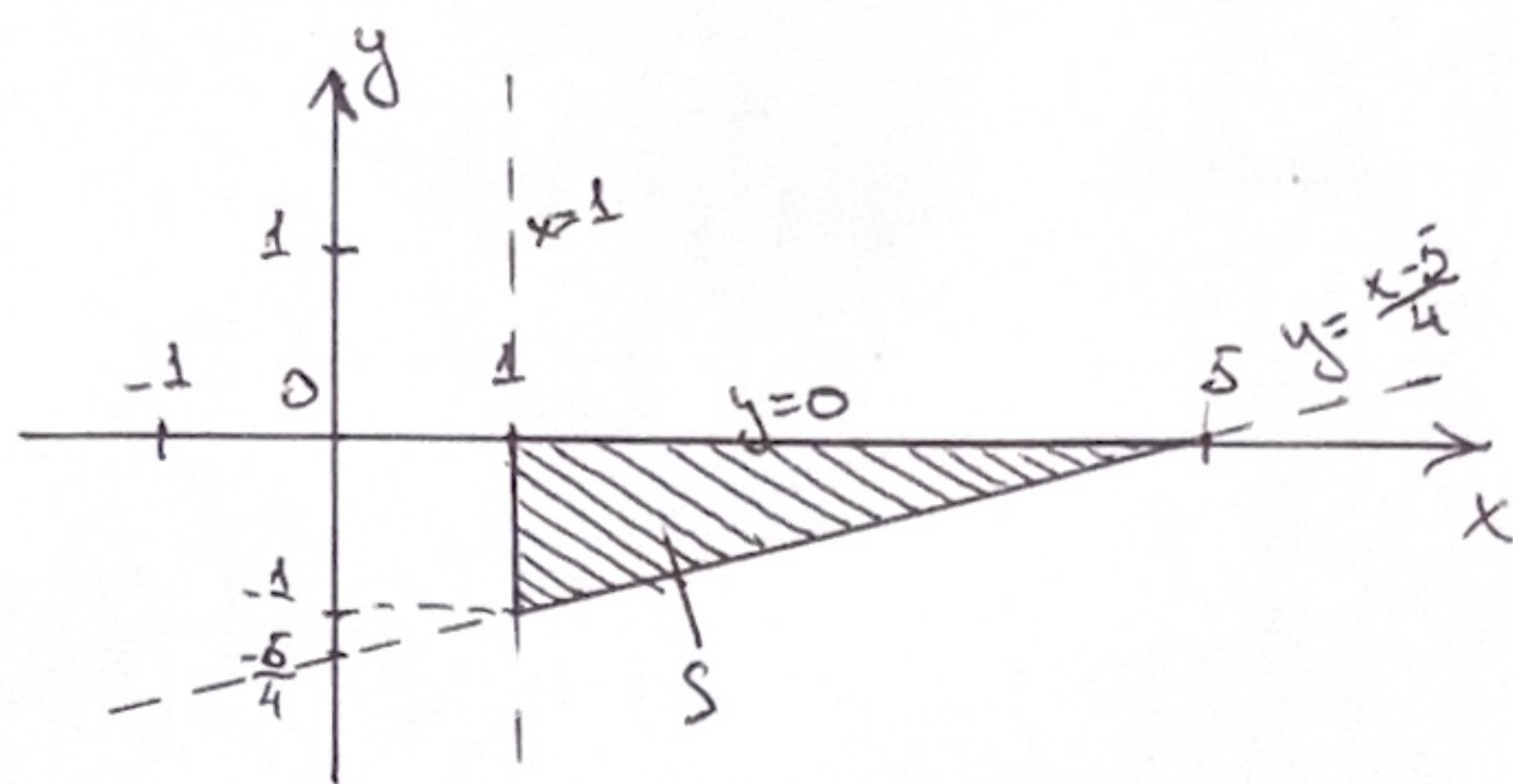
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равномер. распр. случай. вел. (ξ, η) в обл.:

$$\begin{cases} x - 4y \leq 5, \\ x \geq 1, y \leq 0 \end{cases}$$

$$\xi = \xi^* + 1, \quad \eta = [4\eta], \quad \mu = \xi - 4\eta$$



$$1) \quad P_{\xi, \eta}(x, y) = \begin{cases} C, & x \in S \\ 0, & \text{в ост. случ.} \end{cases}$$

$$\iint_S P_{\xi, \eta}(x, y) dx dy = \iint_S C dx dy = C \iint_S dx dy = C \cdot \frac{4 \cdot 1}{2} = 2C \Rightarrow \begin{cases} 2C = 1 \\ C = \frac{1}{2} \end{cases}$$

$$P_{\xi, \eta}(x, y) = \begin{cases} \frac{1}{2}, & x \in S \\ 0, & \text{в ост. случ.} \end{cases}$$

$$P_{\xi}(x) = \int_{\frac{x-5}{4}}^0 \frac{1}{2} dy = \frac{1}{2} \left(0 - \frac{x-5}{4} \right) = \frac{1}{2} \cdot \frac{5-x}{4} = \frac{5-x}{8}, \quad x \in [1; 5]$$

$$P_{\eta}(y) = \int_1^{5+4y} \frac{1}{2} dx = \frac{1}{2} (5+4y - 1) = \frac{1}{2} (4y+4) = 2y+2, \quad y \in [-1; 0]$$

$$P_{\xi}(x) = \begin{cases} \frac{5-x}{8}, & x \in [1; 5] \\ 0, & \text{в ост. случ.} \end{cases}$$

$$P_{\eta}(y) = \begin{cases} 2y+2, & y \in [-1; 0] \\ 0, & \text{в ост. случ.} \end{cases}$$

$$F_{\xi}(x) = \int_1^x \frac{5-t}{8} dt = \frac{1}{8} \int_1^x (5-t) dt = \frac{1}{8} \left(5t - \frac{t^2}{2} \right) \Big|_1^x = \frac{1}{8} \left(5x - \frac{x^2}{2} - 5 + \frac{1}{2} \right) =$$

$$= \frac{1}{8} \left(-\frac{x^2}{2} + 10x - 9 \right) = \frac{-x^2 + 10x - 9}{16}, \quad x \in [1; 5]$$

$$F_{\eta}(y) = \int_{-1}^y 2(1+t) dt = 2 \int_{-1}^y (1+t) dt = 2 \left(t + \frac{t^2}{2} \right) \Big|_{-1}^y = 2 \left(y + \frac{y^2}{2} + 1 - \frac{1}{2} \right) =$$

$$= 2 \left(\frac{y^2 + 2y + 1}{2} \right) = y^2 + 2y + 1 = (y+1)^2, \quad y \in [-1; 0]$$

$$F_{\xi}(x) = \begin{cases} 0, & x < 1 \\ \frac{-x^2 + 10x - 9}{16}, & x \in [1; 5] \\ 1, & x > 5 \end{cases}$$

$$F_{\eta}(y) = \begin{cases} 0, & y < -1 \\ (y+1)^2, & y \in [-1; 0] \\ 1, & y > 0 \end{cases}$$

Проверим независимость компонент:

$$P_{Y,H}(x,y) \neq P_Y(x) \cdot P_H(y)$$

$$\text{I } x=1, y=0$$

$$P_Y(1) = \frac{5-1}{8} = \frac{1}{2}, \quad P_H(0) = 2 \cdot 0 + 2 = 2$$

$$P_{Y,H}(1,0) = \frac{1}{2}$$

$$\frac{1}{2} \neq \frac{1}{2} \cdot 2 \Rightarrow P_Y(x) \text{ и } P_H(y) \text{ зависимы (какой вопрос?)}$$

$$2) \text{ a) } f = Y^3 + 1$$

$$\text{supp}(Y) = [1, 5]$$

$$\text{supp}(f) = [2, 126]$$

$$F_f(x) = P(f < x) = P(Y^3 + 1 < x) = P(Y^3 < x-1) = P(Y < \sqrt[3]{x-1}) =$$

$$= F_Y(\sqrt[3]{x-1}) = \frac{-(x-1)^{2/3} + 10(x-1)^{1/3} - 9}{16}, \quad x \in [2, 126]$$

т.к. $Y > 0$, учт. верх. гран.

$$F_f(x) = \begin{cases} 0, & x \leq 2 \\ \frac{-(x-1)^{2/3} + 10(x-1)^{1/3} - 9}{16}, & x \in [2, 126] \\ 1, & x > 126 \end{cases}$$

$$P_f(x) = \begin{cases} \frac{1}{12} \cdot \frac{5 - (x-1)^{1/3}}{(x-1)^{2/3}}, & x \in [2, 126] \\ 0, & \text{в ост. случ.} \end{cases}$$

$$Ef = \int_{-\infty}^{\infty} x \cdot P_f(x) dx = \int_2^{126} x \cdot \left(\frac{1}{12} \cdot \frac{5 - (x-1)^{1/3}}{(x-1)^{2/3}} \right) dx = \frac{1}{12} \int_2^{126} \frac{5x}{(x-1)^{2/3}} dx - \frac{1}{12} \int_2^{126} \frac{x}{(x-1)^{1/3}} dx = 20,4$$

$$Ef^2 = \int_{-\infty}^{\infty} x^2 \cdot P_f(x) dx = \int_2^{126} x^2 \cdot \left(\frac{1}{12} \cdot \frac{5 - (x-1)^{1/3}}{(x-1)^{2/3}} \right) dx = \frac{1}{12} \int_2^{126} \frac{5x^2}{(x-1)^{2/3}} dx - \frac{1}{12} \int_2^{126} \frac{x^2}{(x-1)^{1/3}} dx = 911,657$$

$$Df = Ef^2 - (Ef)^2 = 911,657 - (20,4)^2 = 495,497$$

$$8) \text{ } v = [4h]$$

$$\text{supp } h = [-1; 0]$$

$$\text{supp } v = \{-4; -3, -2, -1, 0\}$$

$$P(v=a) = P(a \leq v \leq a+1) = P(a \leq 4h \leq a+1) = P\left(\frac{a}{4} \leq h \leq \frac{a+1}{4}\right) =$$

$$= F_h\left(\frac{a+1}{4}\right) - F_h\left(\frac{a}{4}\right) = \left(\left(\frac{a+1}{4}\right) + 1\right)^2 - \left(\frac{a}{4} + 1\right)^2 = \frac{2a+9}{16}$$

$$F_v(a) = \begin{cases} 0, & a \leq -4 \\ \sum_{a=-4}^a \frac{2a+9}{16}, & a \in [-4; 0] \\ 1, & a > 0 \end{cases}$$

$$Ev = \sum_{a=-4}^{-1} \left(a \cdot \frac{2a+9}{16} \right) = -\frac{30}{16}$$

$$Ev^2 = \sum_{a=-4}^{-1} \left(a^2 \cdot \frac{2a+9}{16} \right) = \frac{70}{16}$$

$$Dv = Ev^2 - (Ev)^2 = \frac{70}{16} - \left(-\frac{30}{16}\right)^2 = \frac{55}{64}$$

$$3) EY = \int_{-\infty}^{\infty} x \cdot p_Y(x) dx = \int_1^5 x \cdot \frac{5-x}{8} dx = \frac{1}{8} \left(\int_1^5 5x dx - \int_1^5 x^2 dx \right) = \frac{1}{8} \left(5 \cdot \frac{x^2}{2} \Big|_1^5 - \frac{x^3}{3} \Big|_1^5 \right) =$$

$$= \frac{1}{8} \left(\frac{5}{2} \cdot 24 + \frac{1}{3} \cdot 124 \right) = \frac{7}{3}$$

$$EY^2 = \int_{-\infty}^{\infty} x^2 \cdot p_Y(x) dx = \int_1^5 x^2 \cdot \frac{5-x}{8} dx = \frac{1}{8} \left(5 \int_1^5 x^2 dx - \int_1^5 x^3 dx \right) = \frac{1}{8} \left(5 \frac{x^3}{3} \Big|_1^5 - \frac{x^4}{4} \Big|_1^5 \right) = \frac{19}{3}$$

$$DY = EY^2 - (EY)^2 = \frac{19}{3} - \left(\frac{7}{3}\right)^2 = \frac{8}{9}$$

$$Eh = \int_{-\infty}^{\infty} y \cdot p_h(y) dy = \int_{-1}^0 y \cdot (2y+2) dy = 2 \left(\int_{-1}^0 y^2 dy + \int_{-1}^0 y dy \right) = 2 \left(\frac{y^3}{3} \Big|_{-1}^0 + \frac{y^2}{2} \Big|_{-1}^0 \right) = -\frac{1}{3}$$

$$Eh^2 = \int_{-\infty}^{\infty} y^2 \cdot p_h(y) dy = \int_{-1}^0 y^2 (2y+2) dy = 2 \left(\int_{-1}^0 y^3 dy + \int_{-1}^0 y^2 dy \right) = 2 \left(\frac{y^4}{4} \Big|_{-1}^0 + \frac{y^3}{3} \Big|_{-1}^0 \right) = \frac{1}{6}$$

$$Dh = Eh^2 - (Eh)^2 = \frac{1}{6} - \left(-\frac{1}{3}\right)^2 = \frac{1}{18}$$

$$E \begin{pmatrix} Y \\ h \end{pmatrix} = \begin{pmatrix} \frac{7}{3} \\ -\frac{1}{3} \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

$$EYh = \iint_{\mathbb{R}^2} xy p_{Yh}(x,y) dx dy = \int_1^5 dx \int_{\frac{x-5}{4}}^0 xy \cdot \frac{1}{2} dy = \int_1^5 \frac{1}{2} x \cdot \frac{y^2}{2} \Big|_{\frac{x-5}{4}}^0 dx = \frac{1}{4} \int_1^5 x \left(-\frac{(x-5)^2}{16} \right) dx =$$

$$= -\frac{1}{64} \int_1^5 (x^3 - 10x^2 + 25x) dx = -\frac{2}{3}$$

$$\text{cov}(Y, h) = EYh - EY \cdot Eh = -\frac{2}{3} - \frac{7}{3} \cdot \left(-\frac{1}{3}\right) = \frac{1}{9}$$

$$\rho(Y, h) = \frac{\text{cov}(Y, h)}{\sqrt{DY \cdot Dh}} = \frac{1}{9} : \sqrt{\frac{8}{9} \cdot \frac{1}{18}} = \frac{1}{2}$$

$$\text{var} \begin{pmatrix} Y \\ h \end{pmatrix} = \begin{pmatrix} DY & \text{cov}(Y, h) \\ \text{cov}(Y, h) & Dh \end{pmatrix} = \begin{pmatrix} \frac{8}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{18} \end{pmatrix}$$

$$p_{Y|h}(x) = \frac{p_{Yh}(x,y)}{p_h(y)} = \frac{\frac{1}{2}}{2y+2} = \frac{1}{4y+4}, \quad x \in [1; 5+4y]$$

0, в ост. envy.

$$E(Y|h=y) = \int_1^{5+4y} x \cdot \frac{1}{4y+4} dx = \frac{1}{4y+4} \int_1^{5+4y} x dx = \frac{1}{4y+4} \cdot \frac{x^2}{2} \Big|_1^{5+4y} = \frac{1}{8y+8} ((5+4y)^2 - 1) = 3+2y$$

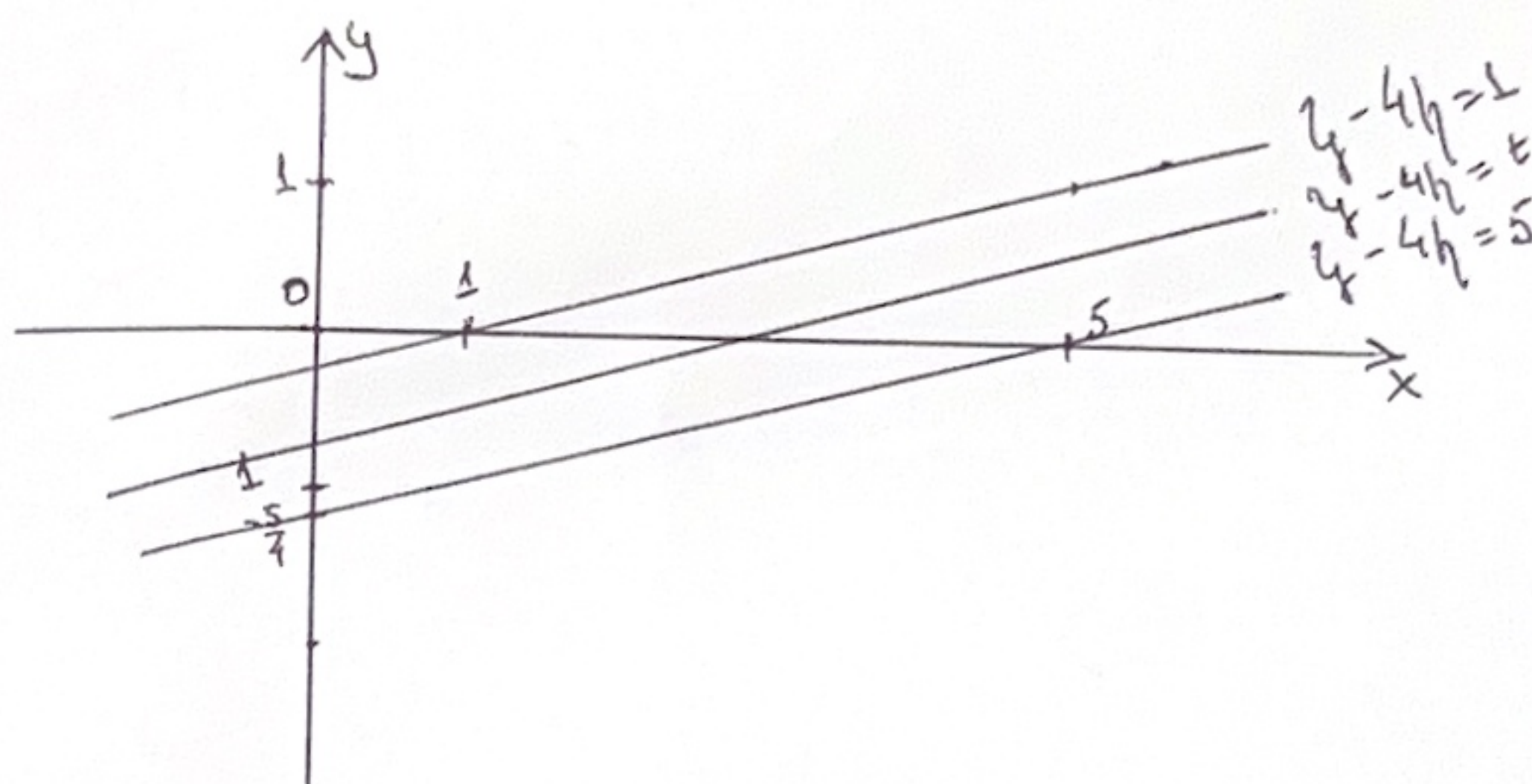
$$E(Y|h) = 2h+3$$

$$E(Y^2|h=y) = \int_1^{5+4y} x^2 \cdot \frac{1}{4y+4} dx = \frac{1}{4y+4} \cdot \frac{x^3}{3} \Big|_1^{5+4y} = \frac{1}{12y+12} ((5+4y)^3 - 1) = \frac{16}{3}y^2 + \frac{44}{3}y + \frac{31}{3}$$

$$D(Y|h=y) = E(Y^2|h=y) - (E(Y|h=y))^2 = \frac{16}{3}y^2 + \frac{44}{3}y + \frac{31}{3} - (2y+3)^2 = \frac{4}{3}y^2 + \frac{8}{3}y + \frac{4}{3}$$

$$D(Y|h) = \frac{4}{3}h^2 + \frac{8}{3}h + \frac{4}{3}$$

$$\begin{aligned}
 4) \quad & y = z - 4h \\
 & \text{supp } z = [1; 5] \\
 & \text{supp } h = [-2; 0] \\
 & z - 4h \leq 5 \text{ (uz onp.)} \\
 & z - 4h = t \\
 & 1 - 4 \cdot 0 = 1 \\
 & 1 - 4 \cdot (-1) = 5 \\
 & \text{supp } y = [1; 5]
 \end{aligned}$$



$$F_Y(t) = P(Y \leq t) = P(z - 4h \leq t)$$

$$F_Y(t) = \begin{cases} 0, & t \leq 1 \\ *, & t \in (1; 5] \\ 1, & t > 5 \end{cases}$$

$$\text{npu } t \in [1; 5]: \quad x - 4y = t \Leftrightarrow y = \frac{x-t}{4}$$

$$y=0 \Rightarrow x=t$$

$$x \geq 1, y \leq 0 \text{ (uz onp.)}$$

$$F_Y(t) = \int_{x=1}^t \int_{y=\frac{x-t}{4}}^0 \frac{1}{2} dy dx = \int_{x=1}^t \frac{1}{2} \left(-\frac{x-t}{4} \right) dx = \frac{1}{8} \int_1^t (t-x) dx = \frac{1}{8} \left(tx - \frac{x^2}{2} \right) \Big|_1^t = \frac{(t-1)^2}{16}$$

$$F_Y(t) = \begin{cases} 0, & t \leq 1 \\ \frac{(t-1)^2}{16}, & t \in (1; 5] \\ 1, & t > 5 \end{cases}$$

$$p_Y(t) = \begin{cases} \frac{t-1}{8}, & t \in (1; 5] \\ 0, & \text{в ост. члгу.} \end{cases}$$

$$E_Y = \int_{-\infty}^{\infty} t p_Y(t) dt = \int_1^5 t \cdot \frac{t-1}{8} dt = \frac{1}{8} \int_1^5 (t^2 - t) dt = \frac{1}{8} \left(\frac{t^3}{3} - \frac{t^2}{2} \right) \Big|_1^5 = \frac{11}{3}$$

$$E_Y^2 = \int_{-\infty}^{\infty} t^2 p_Y(t) dt = \int_1^5 t^2 \cdot \frac{t-1}{8} dt = \frac{1}{8} \int_1^5 (t^3 - t^2) dt = \frac{1}{8} \left(\frac{t^4}{4} - \frac{t^3}{3} \right) \Big|_1^5 = \frac{43}{3}$$

$$D_Y = E_Y^2 - (E_Y)^2 = \frac{43}{3} - \left(\frac{11}{3} \right)^2 = \frac{8}{9}$$